

THE GLAMSQUAD SALON

Data Validation & Dashboard Testing

Portfolio Test Strategy | Data, KPI, Model, Visual & Confidentiality Controls

Document Control	Details
Client	The GlamSquad Salon
Role	Business Analyst + Data Analyst
Project Period	May-Aug 2025
Confirmed Solution	Power BI dashboard built from scratch
Confirmed Sources	Excel and handwritten operational records
Purpose	Define a defensible validation and dashboard-testing approach for the portfolio reconstruction.
Status	Portfolio reconstruction; not a claim of retained historical test scripts or formal UAT sign-off.
Evidence boundary: The project included data analysis and a Power BI dashboard, but exact historical test-case IDs, defect counts, pass/fail logs, UAT approvals and production test evidence are not retained. The test cases below are a professional rebuild specification.	

1. Testing Objectives

- Confirm that prepared data accurately represents the sanitized source records used in the portfolio rebuild.
- Verify that customer, appointment, service, staff, cost and feedback transformations follow documented business rules.
- Reconcile major Power BI KPIs to independently checked source/prepared totals.
- Test model relationships, filters and visual interactions for consistency.
- Prevent unsupported calculations when required denominators or cost attribution are unavailable.
- Verify that public portfolio outputs contain no customer PII or confidential original client values.

2. Testing Scope

Test Layer	In Scope
Source validation	Excel and digitized handwritten-record reconciliation
Data quality	Completeness, duplicates, formats, mappings and invalid values
Transformation	Customer matching, appointment statuses, service/staff standardization, derived fields
Financial reconciliation	Revenue, expenses, profit and conditional service economics
Data model	Keys, relationships and filter propagation
DAX / KPI	Documented measures and denominator handling
Dashboard	Visual totals, filters, cross-highlighting, page consistency and usability
Privacy	PII and confidential commercial-data controls

3. Testing Approach

Sequence: Source checks -> cleaning validation -> transformation checks -> reconciliation -> model validation -> KPI/DAX validation -> visual/filter testing -> privacy review -> management walkthrough.

Testing should compare outputs against an independent control wherever possible. A dashboard visual should not be considered correct merely because it renders without an error.

4. Source & Handwritten-Record Validation

Test ID	Test	Expected Result
DV-01	Confirm each sanitized source has an identified business purpose/grain.	Source purpose and grain are documented.
DV-02	Compare digitized handwritten records with the source during transcription review.	Transcribed values match readable source entries; unclear values are flagged rather than guessed.
DV-03	Check date coverage across prepared sources.	Unexpected gaps/overlaps are investigated or documented.
DV-04	Check key analytical fields for blanks/nulls.	Missing values are quantified in the rebuild and handled according to documented rules.
DV-05	Check numeric fields for invalid text/non-numeric values.	Invalid values are corrected from evidence, flagged or excluded with rationale.
DV-06	Check duplicate candidates.	Potential duplicates are reviewed before deletion/merge.
DV-07	Confirm public data is sanitized before reporting.	No phone numbers or identifiable private customer data remain.

5. Customer & Repeat-Visit Validation

Confirmed business logic: Phone-number data was used to identify repeat visits.		
Test ID	Test	Expected Result
CU-01	Normalize phone-based matching input in the protected preparation layer.	Equivalent formatting does not create artificial separate customers.
CU-02	Review duplicate customer matches.	Ambiguous/shared values are not silently merged.
CU-03	Create sanitized Customer_Key for reporting.	Public model does not require raw phone numbers.
CU-04	Validate Visit_Count against cleaned appointment history.	Count reflects eligible validated visits.
CU-05	Validate Repeat_Customer_Flag.	Flag follows the documented rebuilt repeat rule.
CU-06	Test retention only after a retention window/rule is defined.	No unsupported historical retention rule is implied.

6. Appointment Status Validation

Test ID	Test	Expected Result
AP-01	Map source status variants to standardized categories.	Only evidence-supported statuses map to completed/cancelled/no-show.
AP-02	Keep unresolved statuses separate.	Unknown values are not forced into KPI categories.
AP-03	Reconcile total appointments by standardized status.	Status totals reconcile with eligible prepared records.
AP-04	Validate cancellation-rate denominator.	Rate uses the documented eligible/booked denominator.
AP-05	Validate no-show-rate denominator.	Rate uses the documented eligible/booked denominator.
AP-06	Check completed appointment logic.	Only delivered/attended statuses contribute to completed measures.

7. Service, Pricing & Discount Validation

Test ID	Test	Expected Result
SV-01	Review service-name standardization.	Equivalent service labels map consistently.
SV-02	Reconcile service activity totals.	Service counts sum to the relevant validated appointment/service population.
SV-03	Reconcile Revenue by Service.	Aggregated service revenue matches prepared controls.
SV-04	Validate listed/actual price and discount field meaning in the rebuild.	Amounts and percentages are not mixed.
SV-05	Validate Discount Rate %.	Correct denominator and safe zero/blank handling are used.
SV-06	Validate service contribution/margin prerequisites.	Measure is shown only where direct costs are reliably attributable.
SV-07	Check confidential-value sanitization.	Original service prices/margins are absent from public portfolio data.

8. Staff, Working-Hours & Utilization Validation

Test ID	Test	Expected Result
ST-01	Review staff/stylist label standardization.	One logical sanitized staff identity is used consistently.
ST-02	Reconcile appointments attributed to staff.	Only supported appointment-staff mappings are included.
ST-03	Reconcile Revenue by Stylist.	Attributed revenue matches prepared controls.
ST-04	Validate working-hours values.	Working-hour fields use consistent units and valid values.
ST-05	Validate Revenue per Working Hour.	Both revenue attribution and working-hours denominator are valid.
ST-06	Validate Staff Utilization %.	Displayed only where used/booked time and available capacity are defensible.
ST-07	Validate salary/commission fields.	Numeric period/context is understood and public values are sanitized.
ST-08	Check staff interpretation.	Dashboard does not present revenue alone as a definitive staff-performance score.

9. Revenue, Expense & Profit Reconciliation

Test ID	Test	Expected Result
FN-01	Reconcile Total Revenue to prepared control.	Power BI revenue matches validated prepared data.
FN-02	Reconcile Total Expenses to prepared control.	Power BI expenses match validated prepared data.
FN-03	Validate Profit.	Profit = Revenue - Expenses under identical filter context.
FN-04	Validate Profit Margin %.	Documented formula is used with safe denominator handling.
FN-05	Validate before/after profit comparison where rebuilt.	Same profit definition and comparable reporting context are used.
FN-06	Validate Profit Improvement %.	$(\text{Profit After} - \text{Profit Before}) / \text{Profit Before} \times 100$.
FN-07	Check published outcome wording.	15% profit improvement may be stated; underlying confidential values are not exposed.

10. Feedback Validation

Test ID	Test	Expected Result
FB-01	Verify feedback source/channel classification.	Source labels reflect the confirmed feedback channels where represented.
FB-02	Review theme coding.	Themes are assigned consistently from evidence.
FB-03	Check confirmed themes.	Pricing, staff behaviour, cleanliness and staff availability can be represented.
FB-04	Check privacy.	No identifiable WhatsApp/customer feedback is published.
FB-05	Check quantitative claims.	No invented review counts, sentiment percentages or theme rankings.
FB-06	Check evidence separation.	Qualitative feedback is not aggregated as if it were transactional financial data.

11. Data Model Validation

Test ID	Test	Expected Result
DM-01	Check dimension keys for uniqueness.	Customer, Service, Staff and Date keys are unique at dimension grain.
DM-02	Check fact-to-dimension mappings.	No unintended orphan relationships in required analytical records.
DM-03	Check relationship cardinality.	Logical one-to-many relationships behave as designed.
DM-04	Check filter direction.	Filters propagate intentionally without ambiguous paths.
DM-05	Check Date relationship.	Date filters affect intended fact tables consistently.
DM-06	Check optional cost attribution relationships.	No service/staff cost is assigned without defensible mapping.

12. KPI / DAX Validation Matrix

KPI	Independent Validation
Total Revenue	Sum validated revenue in prepared dataset and compare with measure.
Total Expenses	Sum validated expenses and compare.
Profit	Revenue control minus expense control.
Profit Margin %	Profit divided by Revenue under same filters.
Total Appointments	Count validated appointment grain.
Completed Appointments	Count records mapped to completed/attended logic.
Cancellation Rate %	Validated cancelled count divided by documented eligible denominator.
No-Show Rate %	Validated no-show count divided by documented eligible denominator.
Unique Customers	Distinct sanitized logical customers.
Repeat Customers / Rate	Validate against visit counts and repeat rule.
Revenue by Service	Group prepared revenue by standardized service.
Service Mix %	Validate numerator and documented denominator.
Revenue by Stylist	Group supported attributed revenue by sanitized stylist.
Revenue per Working Hour	Attributed revenue divided by validated working hours.
Staff Utilization %	Used/booked time divided by defensible available capacity.
Service Contribution/Margin	Validate only for records with reliable direct-cost attribution.

13. Dashboard Functional Testing

Test ID	Dashboard Test	Expected Result
DB-01	Open each analytical page.	Page loads without broken visuals.
DB-02	Apply Date filter.	All intended visuals update consistently.
DB-03	Apply Service filter.	Relevant service, revenue and appointment visuals update.
DB-04	Apply Staff/Stylist filter.	Relevant staff and attributed activity/revenue visuals update.
DB-05	Apply Appointment Status filter where present.	Appointment visuals reflect selected status without corrupting unrelated metrics.
DB-06	Test cross-highlighting/cross-filtering.	Interactions are intentional and do not create misleading totals.
DB-07	Clear/reset filters.	Report returns to defined default state.
DB-08	Compare same KPI across pages.	Equivalent filter context returns the same value.
DB-09	Test blank/zero cases.	No divide-by-zero errors or misleading infinity/zero displays.
DB-10	Review labels/titles.	Business user can understand metric and context.

14. Page-Level Dashboard Test Coverage

Page	Critical Tests
Executive Profitability Overview	Revenue/expense/profit reconciliation; profit margin; date filtering; driver consistency; confidential-value control.
Service & Pricing Analysis	Service mapping; revenue by service; service mix; discount logic; conditional contribution/margin.
Staff Performance & Utilization	Staff mapping; attributed revenue/activity; working hours; conditional utilization; salary/commission confidentiality.
Customer & Appointment Analysis	Appointment statuses; cancellations/no-shows; repeat matching; retention-rule boundary; demand-period analysis.

15. Negative & Boundary Testing

- Filter to a period with no records and confirm visuals show blanks/appropriate zero state rather than errors.
- Test a service with no attributable direct cost and confirm margin is not fabricated.
- Test staff records without reliable working-hours capacity and confirm utilization is not calculated.
- Test unknown appointment status and confirm it is not counted as completed/cancelled/no-show.
- Test ambiguous customer matching and confirm records are not silently merged.
- Test zero revenue context and confirm percentage measures avoid divide-by-zero.
- Test feedback with no supported theme and confirm it can remain uncategorized.
- Test public export/screenshot and confirm no customer phone number or original confidential financial value is visible.

16. Traceability to Business Requirements

Requirement Area	Validation Coverage
BR-01 / BR-02 Profitability	FN-01 to FN-07; Executive dashboard tests
BR-03 / BR-04 Appointments	AP-01 to AP-06; DB appointment tests
BR-05 / BR-06 Customers	CU-01 to CU-06
BR-07 / BR-08 / BR-09 Services & Pricing	SV-01 to SV-07
BR-10 to BR-13 Staff / Scheduling / Costs	ST-01 to ST-08
BR-14 Costs	FN tests plus service/staff cost attribution controls
BR-15 Feedback	FB-01 to FB-06
BR-16 Dashboard	DB-01 to DB-10
BR-17 Recommendations	Validated insights must trace to tested metrics/evidence
BR-18 Outcome Monitoring	FN-05 to FN-07

17. Defect Handling — Portfolio Method

Exact historical defect counts and Jira/test logs are not retained. For a portfolio rebuild, any discovered issue should be documented using a simple controlled structure.

Field	Example Purpose
Defect ID	Unique portfolio test issue identifier
Area	Data / KPI / model / visual / privacy
Description	What failed and under which context
Expected Result	Correct business behaviour
Actual Result	Observed behaviour
Severity	Portfolio-defined severity based on business impact
Resolution	Transformation/model/measure/visual correction
Retest Result	Pass/Fail after correction
Do not claim: Do not state historical defect counts, severity distributions, formal defect closure metrics or 'zero defects' for this salon project unless supported by retained evidence.	

18. Management Walkthrough / Acceptance

A live walkthrough is confirmed. It can be described as presenting the analysis/dashboard and recommendations to the manager. Formal UAT sign-off, signed acceptance criteria or a production approval workflow are not confirmed.

Walkthrough Focus	Expected Portfolio Evidence
Business questions	Dashboard pages answer profitability, service, staff and customer/appointment questions.
KPI interpretation	Definitions can be explained in business language.
Findings	Insights trace to validated data/qualitative evidence.
Recommendations	Actions trace to supported root causes.
Confidentiality	No sensitive client/customer data is exposed.

19. Exit Criteria

- Critical data mappings and transformations are validated.
- Revenue, expenses and profit reconcile to prepared controls.
- Appointment, customer, service and staff KPIs pass documented checks.
- Conditional metrics are suppressed when prerequisites are missing.
- Filters and cross-page KPI consistency are tested.
- No customer PII or confidential original client values appear in public outputs.
- Findings used in the case study trace to validated evidence.
- Known portfolio limitations are documented rather than hidden.

20. Confirmed vs Reconstructed Testing Evidence

Classification	Details
Confirmed	Unclean Excel + handwritten source context; Power BI dashboard built from scratch; live walkthrough; analysis covered the documented business areas.
Portfolio reconstruction	Formal test IDs, test matrix, independent reconciliation controls, boundary tests, defect template and exit criteria in this document.
Not retained / not claimed	Exact historical test cases, pass/fail counts, defect counts, Jira records, formal UAT sign-off, production deployment validation, RLS/security testing or exact historical DAX test scripts.